INDOOR AUTONOMOUS DELIVERY ROBOT · REPORT 6 OF 6

Map Evolution: From the First Map to campus_full_map7

Comparative, quantitative review of every ArUco SLAM mapping run in the simulation

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1. What this report compares

Between 4 and 8 November 2026 the ArUco SLAM mapper was run 13 times (plus one empty folder, `campus_full_map2`), first in the two-room `campus_area` world and then in the larger `two_buildings` campus. Each run left a complete map folder in `~/ros2_jazzy_ws/maps`. Because the mapper starts at the Gazebo spawn pose, map coordinates equal world coordinates, so every map can be scored objectively against the marker positions in the world files (`aruco_markers.yaml`, `aruco_markers_two_buildings.yaml`). This report measures each map the same way and explains which change in the software caused each step.

53 m → 1.4 m	100° → 2.6°	10.2 m → 0.39 m	1007 m → 242 m
mean absolute marker error, first → final	global map rotation	shape error after rigid fit	path driven to finish the map

2. Method

- **Absolute error** – distance between each observed marker in `aruco_landmarks.yaml` and its true pose (markers added by hand with 0 observations are left out).
- **Shape error** – the same after the best 2-D rigid transform (rotation + translation, SVD/Kabsch). It shows whether the map is *internally* right; this is what matters for navigation, because the robot localises on the same map.
- **Global rotation** – the angle of that best-fit transform.
- Coverage (markers found), driven path length and simulated time from `trajectory.csv`. Script: `reports/src/scripts/metrics.py` and `align.py`.

3. All runs

Map	Created	World	Markers	Abs. err (m)	Shape err (m)	Rot. (°)	Path (m)	Sim min
campus	11-04 20:54	campus_area	12/12	53.41	10.17	100.4	1007	62.5
campus2	11-04 22:44	campus_area	2/12	1.03	–	–	0	0.2
campus3	11-04 22:52	campus_area	2/12	0.90	–	–	0	0.2
campus4	11-04 23:13	campus_area	5/12	0.34	0.18	1.3	221	14.5
campus5	11-05 00:01	campus_area	11/12	0.63	0.32	0.8	742	44.1
campus6	11-05 00:38	campus_area	12/12	1.61	0.37	2.4	491	29.6
campus_full_map	11-05 04:08	two_buildings	31/32	4.81	0.24	9.1	467	32.6
campus_full_map1	11-07 18:41	two_buildings	0/32	–	–	–	0	0.0
campus_full_map3	11-07 19:44	two_buildings	30/32	4.25	0.66	7.7	464	30.9
campus_full_map4	11-07 21:18	two_buildings	32/32	1.81	1.34	0.7	489	32.8
campus_full_map5	11-07 22:02	two_buildings	11/32	2.20	0.59	1.9	129	10.2
campus_full_map6	11-07 23:39	two_buildings	0/32	–	–	–	1	1.0
campus_full_map7	11-08 00:16	two_buildings	19/32	1.35	0.39	2.6	242	20.9

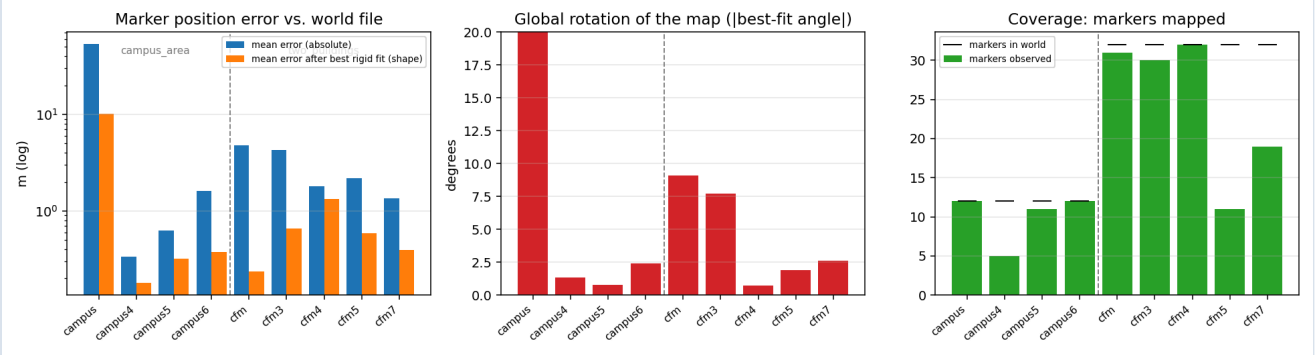


Figure 1. Error, rotation and coverage of every complete or partial run (aborted runs left out). The dashed line separates the two worlds. Absolute error (log scale) is dominated by rotation; the shape error shows the internal quality.

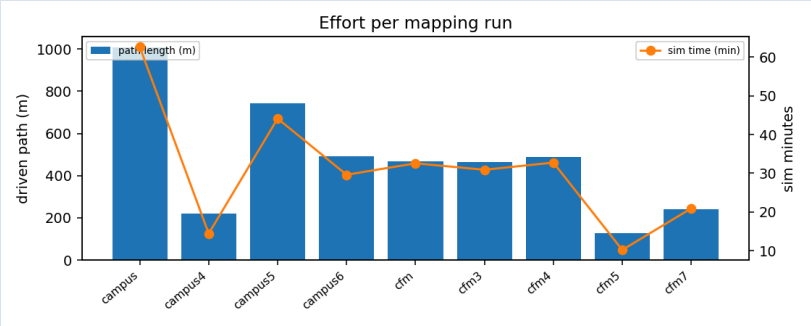


Figure 2. Driven path and simulated time per run. The final map needed the least driving of all full two_buildings runs.

4. Stage 1 – campus_area (4–5 November)

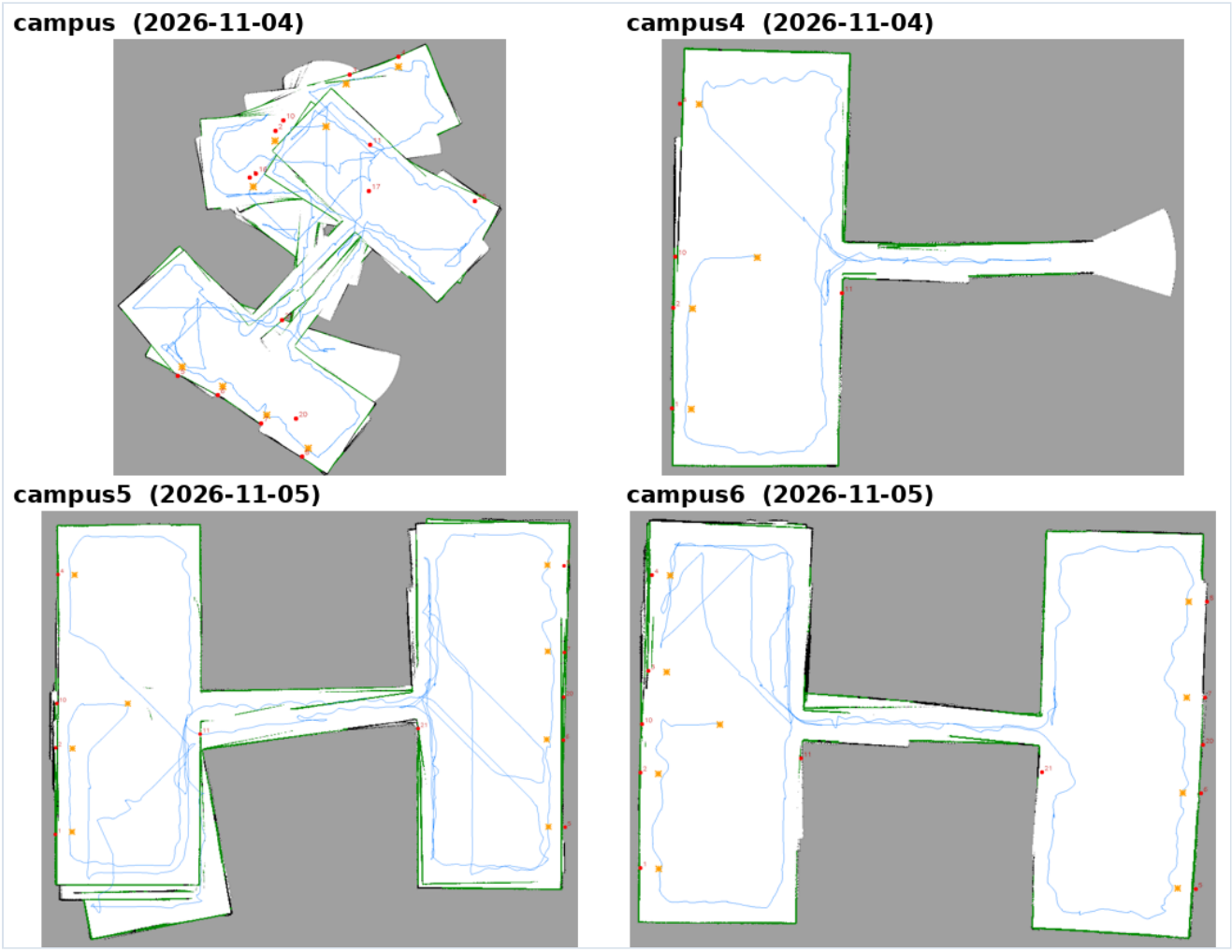


Figure 3. campus_area maps in order. Green walls were seen by the camera, red dots are markers, stars delivery points, blue the driven path.

campus. First full autonomous run. All 12 markers found, but heading from slipping wheels in in-place turns made each pass of a room come out rotated: overlapping copies of both rooms.

campus2. Aborted after 14 s (test of the new scan recording, `scans.npz`).

campus3. Aborted after 14 s (same test).

campus4. Partial run (left room + corridor) after switching heading to the IMU gyro: first clean, straight geometry.

campus5. Both rooms, 11/12 markers. Walls in the right room are doubled (small residual heading error). Live and post-corrected versions saved.

campus6. 12/12 markers, single clean walls in both rooms. Reference map for campus_area.

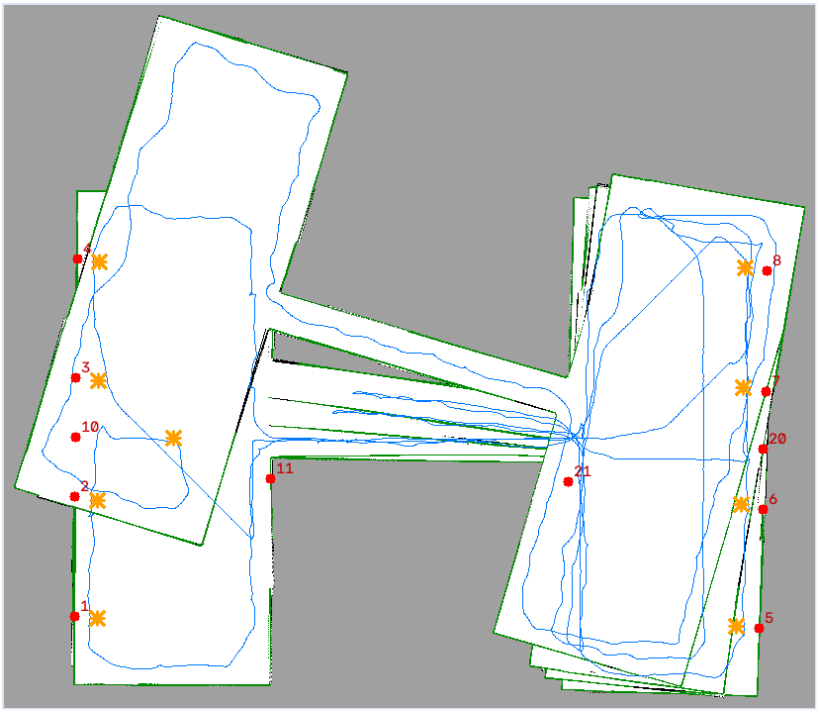


Figure 4. Offline replica test with 15 % rotational wheel slip: the same failure as the first `campus` map – each room is drawn several times at different angles. This test led to using the gyro for heading and to the slip correction tools.

> **What changed:** heading from the IMU gyro instead of the wheels (`use_imu`), slower turns while mapping (0.45 rad/s), and recording raw scans for offline correction. Result: from a shattered map (53 m mean error) to campus6 with all markers and 0.37 m shape error.

5. Stage 2 – two_buildings (5–8 November)

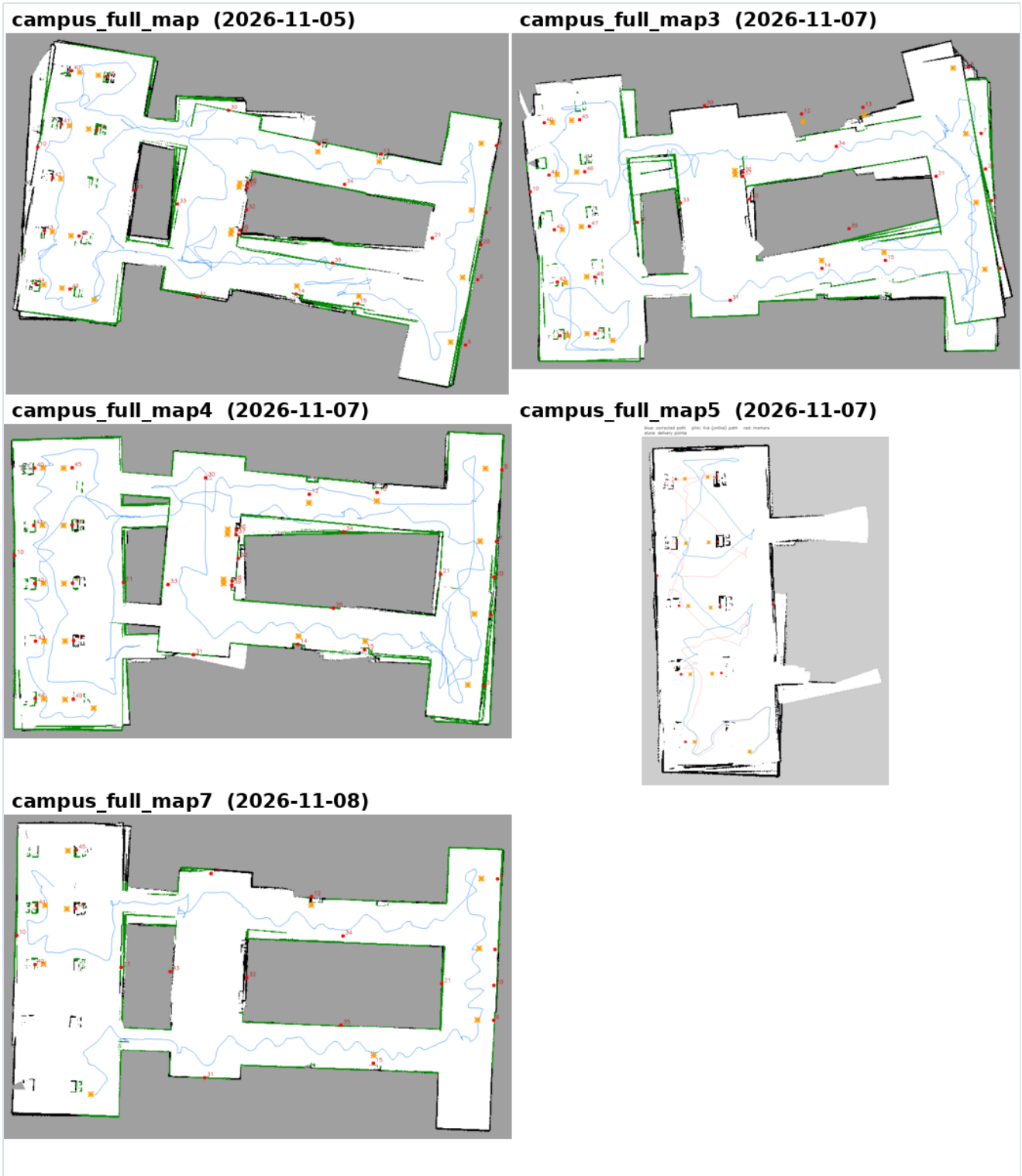


Figure 5. two_buildings maps in order: campus_full_map, 3, 4, 5 and the final 7.

campus_full_map. First map of the larger two_buildings world (32 markers, tables, ring corridor). Shape good, but the whole map is rotated ≈9°.

campus_full_map1. Aborted at start (no trajectory).

campus_full_map3. Rotated ≈8° with ghost walls in building B: marker re-sightings were still allowed to turn the heading.

campus_full_map4. After the "spin fix" (gyro trusted, markers correct position only): rotation gone (0.7°), all 32 markers, but local distortions remain (1.3 m after rigid fit).

campus_full_map5. Building A only, used to develop the post-run pose-graph correction (online vs corrected path: mean 1.78 m apart).

campus_full_map6. Aborted after 61 s.

campus_full_map7. Final map: single sharp walls everywhere, 19 markers observed + 10 table markers added afterwards, shortest run (242 m). Used by the web app.

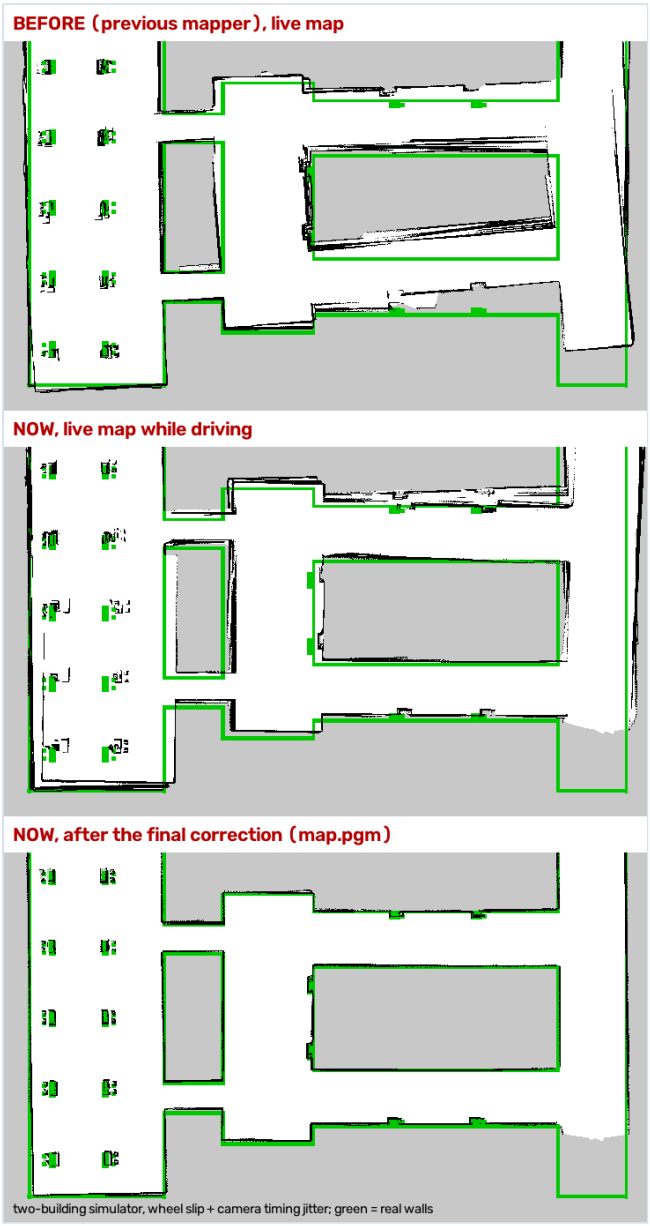


Figure 6. The "spin fix": previous mapper (top), live map while driving with the fix (middle), and after the final correction (bottom).

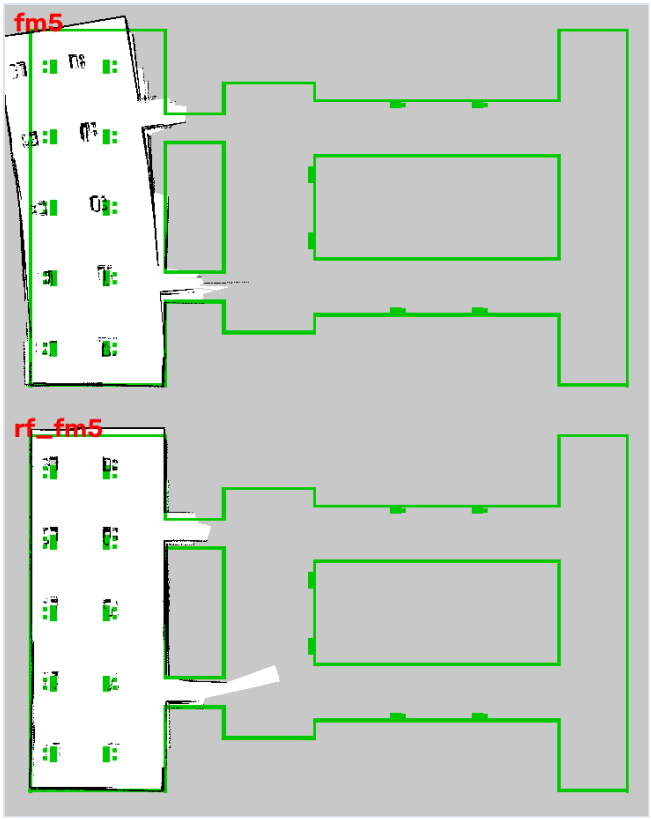


Figure 7. campus_full_map5 live (top, bent building A) and re-processed with the fixed mapper (bottom) against the true walls (green).

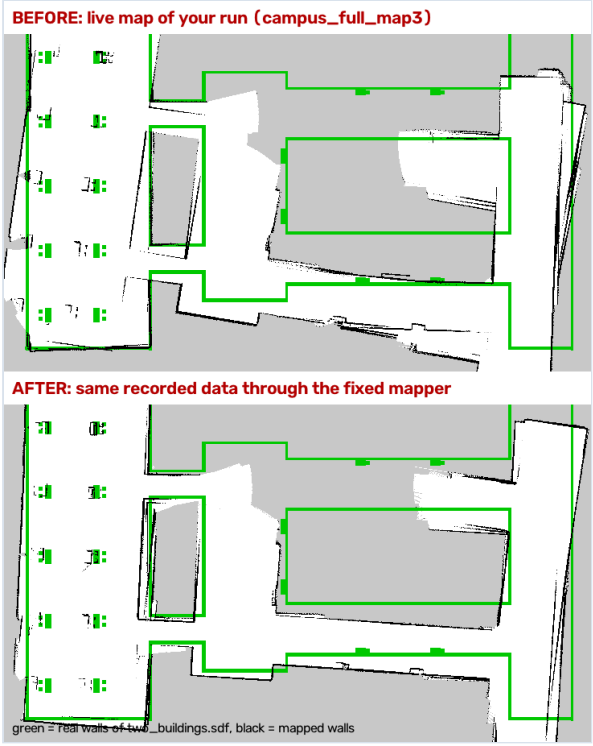


Figure 8. campus_full_map3: live map (top) and the same recorded data through the fixed mapper (bottom).

> **What changed:** (1) the gyro is trusted strongly in the EKF (`gyro_noise 0.0003`) so a marker seen again corrects the *position* but can no longer rotate the map; (2) marker bearings are down-weighted while turning and ignored above 0.5 rad/s; (3) sensor time stamps are synchronised with the motion (`sync_sensors`) for runs faster than real time; (4) exploration targets walls the camera has not yet seen from close range, so fewer passes are needed. The rotation fell from 9° to under 3°, and the final run drove 242 m instead of ≈470–490 m.

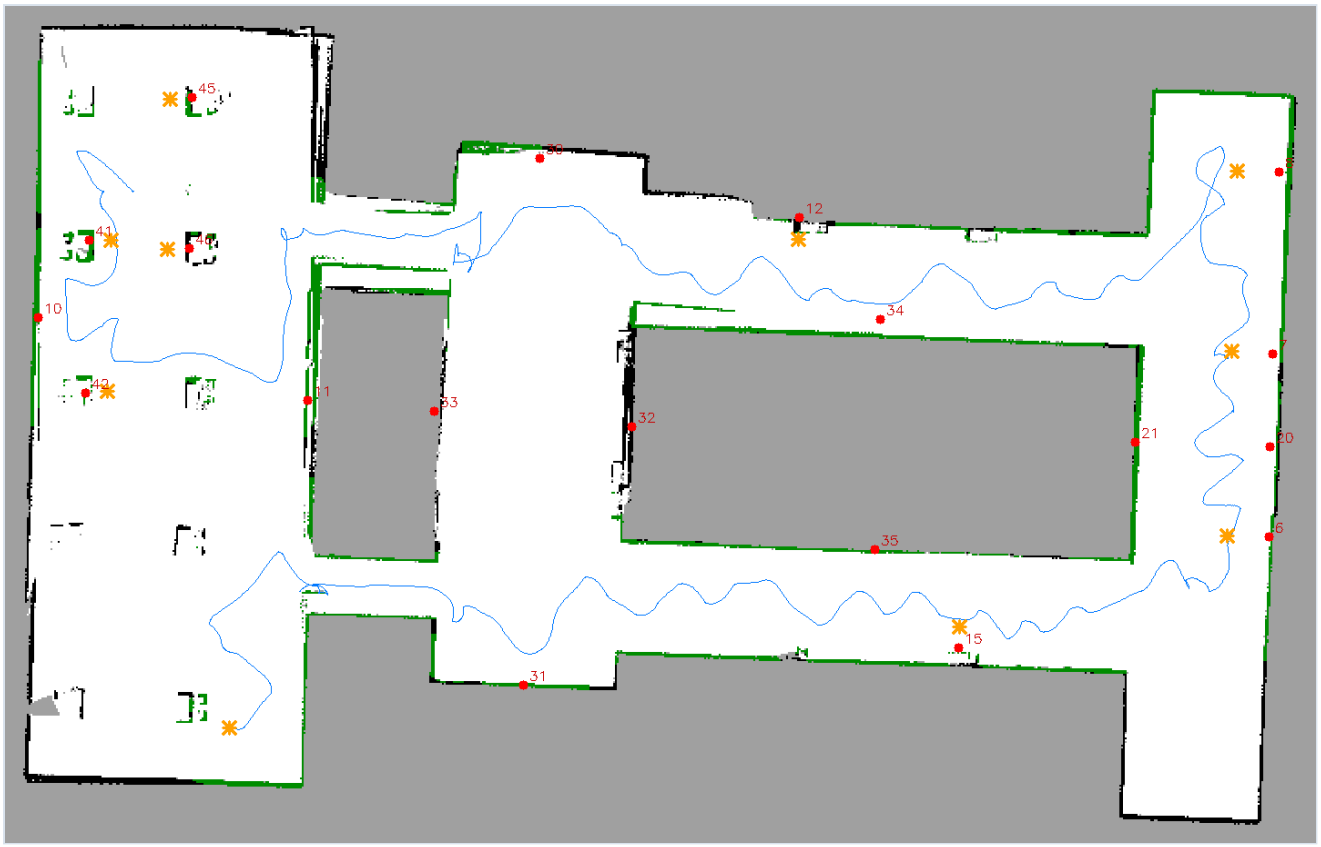


Figure 9. The final map campus_full_map7 used by the web application and the delivery navigator.

6. First vs. last map

	campus (first)	campus6 (best campus_area)	campus_full_map (first two_buildings)	campus_full_map7 (final)
Markers observed	12/12	12/12	31/32	19/32 (+10 added)
Mean absolute error	53.4 m	1.61 m	4.81 m	1.35 m
Max absolute error	114.1 m	2.90 m	9.68 m	1.89 m
Shape error (rigid fit)	10.2 m	0.37 m	0.24 m	0.39 m
Global rotation	100°	2.4°	9.1°	2.6°
Walls	multiple rotated copies	single, straight	single, slightly doubled in B	single, sharp
Path / sim time	1007 m / 63 min	491 m / 30 min	467 m / 33 min	242 m / 21 min
Usable for delivery	no	yes	yes, rotated	yes – in production

7. Observations

- **Heading was the dominant error source**, not marker detection. Every big improvement came from how heading is estimated (gyro instead of wheels, gyro over marker bearings).
- **Absolute error and usefulness differ.** campus_full_map has a smaller shape error (0.24 m) than map7 (0.39 m) but is rotated 9°; map7 has sharper walls, which matters for lidar localisation and planning through doors.
- **Coverage vs. marker count.** map7 observed 19 of 32 markers directly: the table markers face inward and were added afterwards from the table layout (`aruco_landmarks_before_tables.yaml` holds the observed-only version). Markers 5, 44 and 49 were never detected and could be added in a short teleoperated top-up run.
- **Short aborted runs** (campus2/3, map1, map6) were start-up tests of new features and are not maps.

> **Conclusion.** Over 13 runs the map went from unusable (rooms drawn several times, 53 m mean error) to campus_full_map7, a clean map of a 70 × 45 m two-building campus with sub-0.4 m internal consistency, built autonomously in 21 simulated minutes and now used for every delivery in the web application.

> **Next:** a top-up run for the three missing markers; run `optimize_map` on map7 and compare the corrected version with the same metrics; repeat the protocol on the real robot with a tape-measured marker survey as ground truth.